Reference sheet: the `MahlerExperiments` folder

Layout. Each experiment family (`rk`, `v2`, `v3`, `v5`, `v7`) has one subfolder per feature combination, named as in Table 1. Each configuration contains three runs with the standard random split. The `v2` family also contains two blocked splits, with ten runs per split (Table 3). The archive contains 104 runs in total.

Each run folder is named by its run identifier and contains three files:

- `train.log`
  the training log and run configuration.
- `eval.valid.epochs_0_499.json.bz2`
  per-example validation predictions.
- `eval.test1.epochs_0_499.json.bz2`
  per-example test predictions.

The master file `data/output_full.txt` gives r_k as a fraction, together with the conductor and other arithmetic invariants for each k. The run exports provide logs and predictions. Model weights and tokenized training inputs are separate.

Accuracy criteria. Exact match requires equality with the target. Factor-two accuracy for r_k requires the correct sign and a predicted magnitude strictly between half and twice the true magnitude. For the valuation experiments, the stored target is $v_p(r_k)$. Changing its sign gives $v_p(n_k)$, where $n_k = 1/r_k$.

Checkpoint selection. Epochs are numbered from 0. For the principal comparisons, select the earliest epoch attaining the highest validation score among epochs 0-299 and report the corresponding test score. Learning curves use validation predictions.

Notation. All runs take k as input. In Table 1, ω denotes the root number, N the conductor, Δ the discriminant, a_p the Frobenius traces and c_2 the Tamagawa number at 2. The rank feature consists of the rank bounds. Factor denotes prime factorization.

TABLE 1. Explanation of the folder names.

Experiment	Target	Extra features
rk		none
rk_w		ω
rk_wN		ω, N
rk_wap		ω, a_p
rk_wNap	r_k	ω, N, a_p
rk_wfN		ω, factor(N)
rk_wD		ω, Δ
rk_wfD		ω, factor(Δ)
rk_wr		ω, rank
v2		none
v2_N		N
v2_fN		factor(N)
v2_D	$v_2(r_k)$	Δ
v2_fD		factor(Δ)
v2_fDfN		factor(Δ), factor(N)
v2_c2		c_2
v2_fDc2		factor(Δ), c_2
v3		none
v3_N		N
v3_fN	$v_3(r_k)$	factor(N)
v3_D		Δ
v3_fD		factor(Δ)
v3_fDfN		factor(Δ), factor(N)
v5		none
v5_fN	$v_5(r_k)$	factor(N)
v5_fD		factor(Δ)
v7	$v_7(r_k)$	none
v7_fN		factor(N)

TABLE 2. Runs used for the figures.

Figure(s)	Run folder or source
3.1, 3.3, 3.4, 7.1, 7.2	`data/output_full.txt`
3.2	TikZ source in the manuscript
5.1	`rk/rk/1000262_0-2` `rk/rk_w/1000262_3-5`

5.2	`rk/rk_wN/1000262_9-11`
	`rk/rk_wfD/1000262_18-20`
	`rk/rk_wfN/1000262_21-23`

5.3	`rk/rk_wfD/1000262_18-20`
5.4-5.8	`v3/v3_fD/1000260_9-11`
5.9-5.13	`v2/v2_fD/1000261_12-14`

A suffix range denotes separate run directories. For example, `1000262_0-2` denotes `1000262_0`, `1000262_1` and `1000262_2`.

TABLE 3. Blocked splits for the 2-adic experiment.

Run folder	Training k	Validation k	Test k
`v2/reshuffled_end` `1000304_0-9`	1-244,000	244,001-247,000	247,001-250,000
`v2/reshuffled_middle` `1000304_10-19`	1-125,000 and 131,001-250,000	125,001-128,000	128,001-131,000

Both splits use k and the discriminant factorization as inputs. The singular value $k = 4$ is excluded from training.

At epoch 40, mean test accuracy was 23.38% on the upper block and 23.45% on the interior block. At the validation-selected checkpoints, the corresponding values were 37.61% and 37.86%, with sample standard deviations of 0.91 and 1.34 percentage points over ten runs. The explicit predictor achieved 24.83% and 24.03%, respectively, on the same test blocks.